

$$(20) \int \frac{3x+1}{(3x-1)(x+1)} = \frac{A}{3x-1} + \frac{B}{x+1}$$

$$3x+1 = A(x+1) + B(3x-1)$$

$$= x(A+3B) + (A-B)$$

$$A+3B=3 \quad \text{--- (1)}$$

$$A-B=1 \quad \text{--- (2)}$$

$A = \frac{3}{2}$
$B = \frac{1}{2}$

$$\frac{3}{2} \int \frac{1}{3x+1} + \frac{1}{2} \int \frac{1}{x+1} = \frac{3}{2} \ln|3x+1| + \frac{1}{2} \ln|x+1|$$

$$(27) \int \frac{x^2}{(x+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}$$

$$x^2 = A(x+1)^2 + B(x+1) + C \quad \text{--- (1)}$$

$$x^2 = Ax^2 + 2Ax + A + Bx + B + C$$

$$x^2 = Ax^2 + x(2A+B) + (A+B+C) \quad \text{--- (2)}$$

$A = 1$

$$x^2 = A(x+1)^2 + B(x+1) + C$$

$$(-1)^2 = A(-1+1)^2 + B(-1+1) + C$$

$\therefore C = 1$

From (2) $\Rightarrow$

$$x^2 = (x+1) + B(x+1) + 1$$

$$x=0 \Rightarrow$$

$$0^2 = 1 + B + 1$$

$$\therefore B = -2$$

$$= \int \frac{1}{x+1} - \int \frac{2}{(x+1)^2} + \int \frac{1}{(x+1)^3}$$

$$\Rightarrow \ln|x+1| + \frac{2}{x+1} - \frac{1}{2(x+1)^2} + C$$

$$(30) \int \frac{dx}{x(x^2+2)} = \frac{A}{x} + \frac{Bx+C}{x^2+2}$$

$$1 = A(x^2+2) + (Bx+C)x$$

~~$$1 = A(x^2+2) + (Bx+C)x$$~~

$$1 = Ax^2 + 2A + Bx^2 + Cx$$

$$= x^2(A+B) + Cx + 2A$$

$$= 1$$

$$x=0$$

$$1 = A(2) + 0$$

$$\boxed{A = \frac{1}{2}}$$

from (1)

$$A+B=0$$

$$\frac{1}{2} + B = 0$$

$$\boxed{B = -\frac{1}{2}}$$

$$(1) \Rightarrow 1 = x^2\left(\frac{1}{2} - \frac{1}{2}\right) + Cx + 2\left(\frac{1}{2}\right)$$

$$1 = x^2 \times 0 + Cx + 1$$

$$1 = 0 + Cx + 1$$

$$Cx = 0$$

$$\boxed{\therefore C = 0}$$

Now $\Rightarrow \frac{1}{2} \int \frac{x}{x^r+2}$

$$= \frac{1}{2} \ln|x| - \int \frac{x \, dx}{2(x^r+2)}$$

$$= \frac{1}{2} \ln|x| - \frac{1}{2} \times \frac{1}{2} \int \frac{du}{u}$$

$$= \frac{1}{2} \ln|x| - \frac{1}{4} \ln(x^r+2)$$

Let

$$x^r+2=u$$

$$2x \, dx = du$$

$$x \, dx = \frac{du}{2}$$

[31] $\int \frac{x^3 + 3x^r + x + 9}{(x^r+1)(x^r+3)} \equiv \frac{Ax+B}{x^r+1} + \frac{Cx+D}{x^r+2}$

$$= x^3 + 3x^r + x + 9 = Ax + B(x^r+3) + Cx + D(x^r+1)$$

$$= x^3 + 3x^r + x + 9 = Ax^3 + Bx^r + 3Ax + 3B + Cx^3 + Dx^r + Cx + D$$

$$= x^3(A+C) + x^r(B+D) + x(3A+C) + 3B+D$$

$$A+C=1 \quad \text{--- (i)}$$

$$B+D=3 \quad \text{--- (ii)}$$

$$3A+C=1 \quad \text{--- (iii)}$$

$$3B+D=9 \quad \text{--- (iv)}$$

By solving (i) and (iii)

$$\boxed{A=0} \quad \boxed{C=1}$$

(ii) and (iv)

$$\boxed{B=3} \quad \boxed{D=0}$$

$0 + \tan^{-1} x$

Now $\Rightarrow \int \frac{3}{x^r+1} + \int \frac{x}{x^r+2} \Rightarrow 3 \ln|x^r+1| + \int \frac{x}{x^r+2}$

$$\int \frac{x}{x^2+2}$$

Let $u = x^2 + 2$
 $2x dx = du$
 $x dx = \frac{du}{2}$

$$= \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln(x^2+2)$$

So the final solution is $\Rightarrow$

$$= 3 \tan^{-1} x + \frac{1}{2} \ln(x^2+2)$$

Chapter 7.8

③ ~~1.1~~ $\int_0^{+\infty} e^{-2x} dx$

$$\Rightarrow \lim_{t \rightarrow \infty} \int_0^t e^{-2x} dx$$

$$\Rightarrow -\frac{1}{2} \lim_{t \rightarrow \infty} [e^{-2x}]_0^t$$

$$= -\frac{1}{2} \lim_{t \rightarrow \infty} (e^{-2t} - e^0)$$

$$= -\frac{1}{2} (e^{-2 \times \infty} - 1) = -\frac{1}{2} (0 - 1) = \frac{1}{2}$$

⑤ $\int_3^{+\infty} \frac{2}{x^2-1} dx$

$$\Rightarrow \lim_{t \rightarrow \infty} \int_3^t \frac{2}{x^2-1} = 2 \lim_{t \rightarrow \infty} \int_3^t \frac{dx}{x^2-1}$$

$$\frac{2x}{x^2-1}$$

$$= 2 \lim_{t \rightarrow \infty} \left[\cot^{-1} t - \cot^{-1} 3 \right]$$

$$= 2 \lim_{t \rightarrow \infty} (\cot^{-1} t - \cot^{-1} 3)$$

$$\Rightarrow 2 \lim_{t \rightarrow \infty} (\cot^{-1} \infty - \cot^{-1} 3)$$

$$= 2 \cot^{-1} 3 \quad (\text{convergence})$$

$$\boxed{7} \int_e^\infty \frac{dx}{x \ln 3x}$$

$$\Rightarrow \int_e^\infty \frac{du}{u^3}$$

$$= \lim_{t \rightarrow \infty} \left[\frac{u^{-2}}{-2} \right]_e^t$$

$$= -\frac{1}{2} \lim_{t \rightarrow \infty} \left[\frac{1}{(\ln t)^2} \right]_e^t = -\frac{1}{2} \lim_{t \rightarrow \infty} \left(\frac{1}{(\ln t)^2} - \frac{1}{(\ln e)^2} \right)$$

$$= -\frac{1}{2} \lim_{t \rightarrow \infty} \left(\frac{1}{(\ln t)^2} - 1 \right)$$

$$= -\frac{1}{2} \left(\frac{1}{(\ln \infty)^2} - 1 \right) = -\frac{1}{2} (0 - 1) = \frac{1}{2}$$

Let: $\ln x = u$

$$\frac{dx}{x} = du$$

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$$\int_{-\infty}^{\infty} \frac{x}{(x^2+3)^2} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{du}{u^2}$$

Let
 $x^2+3 = u$
 $2x dx = du$
 $x dx = \frac{du}{2}$

$$f(a) + f(b)$$

$$= \frac{1}{2} \lim_{a \rightarrow \infty} \int_0^a \frac{du}{u^2}$$

$$= -\frac{1}{2} \lim_{a \rightarrow \infty} \int_0^a \frac{u^{-1}}{-1} = -\frac{1}{2} \lim_{a \rightarrow \infty} \int_0^a \frac{1}{u}$$

$$= \frac{1}{2} \lim_{a \rightarrow \infty} -[u]_0^a$$

$$= -\frac{1}{2} \lim_{a \rightarrow \infty} [x^2+3]_0^a \Rightarrow -\frac{1}{2} [\infty^2+3 - (0+3)]$$

$$= -\frac{1}{2} (\infty) \left(\frac{1}{\infty} - \frac{1}{3} \right) = -\frac{1}{2} \left(-\frac{1}{3} \right) = \frac{1}{6}$$

This is finite. So the other part -

$$= \frac{1}{2} \lim_{b \rightarrow \infty} \int_{-b}^0 \frac{du}{u^2} = -\frac{1}{2} \lim_{b \rightarrow \infty} \left[\frac{1}{u} \right]_{-b}^0$$

$$= -\frac{1}{2} \lim_{b \rightarrow \infty} \left[\frac{1}{x^2+3} \right]_{-b}^0 = -\frac{1}{2} \left[\frac{1}{0^2+3} - \frac{1}{\infty^2+3} \right]$$

$$= -\frac{1}{6}$$

$$\therefore \text{Final solution} = \frac{1}{6} - \frac{1}{6} = 0$$

This is convergent.

(23)

$$\int_{\pi/3}^{\pi/2} \frac{\sin x}{\sqrt{1-2\cos x}} = \lim_{b \rightarrow \pi/3} \int_b^{\pi/2} \frac{\sin x}{\sqrt{1-2\cos x}}$$

$$= \frac{1}{2} \lim_{b \rightarrow \pi/3} \int_b^{\pi/2} \frac{du}{\sqrt{u}}$$

$$= \frac{1}{2} \lim_{b \rightarrow \pi/3} \int_b^{\pi/2} \frac{du}{2\sqrt{u}}$$

$$= \lim_{b \rightarrow \pi/3} \left[\sqrt{u} \right]_b^{\pi/2} = \lim_{b \rightarrow \pi/3} \left[1 - 2\cos x \right]_b^{\pi/2}$$

$$= (1 - 2\cos \pi/2) - (1 - 2\cos b)$$

$$= 1 - (1 - 2\cos \pi/3) = 1 - (1 - 1) = 1 - 0 = 1$$

This is convergent.

$$\begin{aligned} 1 - 2\cos x &= u \\ 2\sin x dx &= du \\ \sin x dx &= \frac{du}{2} \end{aligned}$$

$$\int_0^{\infty} \frac{1}{x^2} dx = \lim_{a \rightarrow \infty} \int_0^a \frac{1}{x^2} dx = -\lim_{a \rightarrow \infty} \left[\frac{1}{x} \right]_0^a$$

$$= -\lim_{a \rightarrow \infty} \left(\frac{1}{a} - \frac{1}{0} \right) = -\left(\frac{1}{\infty} - \infty \right) = -(0 - \infty)$$

$$= \infty \text{ [divergent]}.$$

7.3

[19] $\int_0^{\pi/3} \sin^4 3x \cos^3 3x dx =$

$$= \int_0^{\pi/3} \sin^4 3x (1 - \sin^2 3x) \cos 3x dx$$

$$= \left[\frac{1}{15} \sin^5 3x - \frac{1}{21} \sin^7 3x \right]_0^{\pi/3}$$

$$= \left[\frac{1}{15} (\sin \pi)^5 - \frac{1}{21} (\sin \pi)^7 - \frac{1}{15} (\sin 0)^5 + \frac{1}{21} (\sin 0)^7 \right]$$

$$= 0 - 0 - 0 + 0 = 0$$

[43] $\int \sqrt{\tan x} \sec^4 x dx$

$$= \int \sqrt{\tan x} (1 + \tan^2 x) \sec^2 x dx$$

$$= \int \sqrt{u} (1 + u^2) du$$

Let
 $\tan x = u$
 $\sec^2 x dx = du$

$$\int (\sqrt{u} + u^{5/2}) du = \frac{2}{3} u^{3/2} + \frac{2}{7} u^{7/2}$$

$$= \frac{2}{3} (\tan x)^{3/2} + \frac{2}{7} \tan^{7/2} x + C.$$

A7 $\int_0^{\pi/2} \tan^5 \frac{x}{2} dx$

$$\frac{x}{2} = y$$

$$\frac{dx}{2} = dy$$

$$dx = 2dy$$

$$\Rightarrow 2 \int \tan^5 u du$$

$$\Rightarrow 2 \left[\frac{\tan^6 u}{6} \right]_0^{\pi/2}$$

$$= \frac{1}{3} \left[(\tan \frac{\pi}{2})^6 - (\tan 0)^6 \right]$$

$$= 2 \int \tan^3 x \tan^2 x dx = 2 \int \tan^3 x (\sec^2 x - 1) dx$$

$$= 2 \int \tan^3 x \sec^2 x dx - 2 \int \tan^3 x dx$$

$$\Rightarrow 2 \int p^3 dp - 2 \int p^2 \tan x (\sec^2 x + 1) dx$$

Let
 $\tan x = p$
 $\sec^2 x dx = dp$

$$= 2 \int p^3 dp - 2 \int p^2 (\tan x \sec^2 x + \tan x) dx$$

$$= 2 \frac{p^4}{4} - 2 \int p^2 dp - 2 \int p dp$$

$$= \frac{p^4}{2} - \frac{1}{4} \quad \frac{1}{2} \quad \frac{1}{2} - \tan^{-1} u - \tan^{-1} u$$

$$= \frac{\tan^4 \left[\frac{x}{2} \right] - \tan^2 \left[\frac{x}{2} \right] - \tan \left[\frac{x}{2} \right] + C}{2}$$

6.4

⑤ $y = x^{2/3}$

$$\frac{dy}{dx} = \frac{2}{3} x^{-1/3}$$

$$\left(\frac{dy}{dx} \right)^2 = \left(\frac{2}{3} x^{-1/3} \right)^2 = \frac{4}{9} x^{-2/3}$$

$$1 + \left(\frac{dy}{dx} \right)^2 = 1 + \frac{4}{9} x^{-2/3} = \frac{9 + 4x^{-2/3}}{9}$$

$$= \frac{9x^{2/3} + 4}{9x^{2/3}}$$

Arc Length

$$L = \int_1^8 \frac{\sqrt{9x^{2/3} + 4}}{9x^{2/3}} dx$$

$$= \int_1^8 (9x^{2/3} + 4)^{1/2} \cdot \frac{1}{9} x^{-2/3} dx$$

$$\begin{aligned}
 &= \frac{1}{18} \int_{13}^{40} u^{1/2} du \\
 &= \frac{1}{27} \left[u^{3/2} \right]_{13}^{40} \\
 &= \frac{1}{27} (40^{3/2} - 13^{3/2}) \\
 &= \frac{1}{27} (40\sqrt{40} - 13\sqrt{13})
 \end{aligned}$$

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$$x = \cos 2t, \quad y = \sin 2t \quad (0 \leq t \leq \pi/2)$$

$$\frac{dx}{dt} = -2 \sin 2t$$

$$\frac{dy}{dt} = 2 \cos 2t$$

$$\left(\frac{dx}{dt} \right)^2 + \left(\frac{dy}{dt} \right)^2 = (-2 \sin 2t)^2 + (2 \cos 2t)^2$$

$$= 4 \sin^2 2t + 4 \cos^2 2t$$

$$= 4(\sin^2 2t + \cos^2 2t)$$

$$= 4(1)$$

$$L = \int_0^{\pi/2} 2 dt = \left[2t \right]_0^{\pi/2} = 2(\pi/2 - 0)$$

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$$x = e^t(\sin t + \cos t) \quad y = e^t(\cos t - \sin t)$$

$$\frac{dx}{dt} = e^t(\sin t + \cos t) + e^t(\cos t - \sin t)$$

$$= 2e^t \cos t$$

$$\frac{dy}{dt} = e^t(\cos t - \sin t) + e^t(-\sin t - \cos t)$$

$$= -2e^t \sin t$$

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (2e^t \cos t)^2 + (-2e^t \sin t)^2$$

$$= 4e^{4t} \cos^2 t + 4e^{4t} \sin^2 t$$

$$= 4e^{4t} (\cos^2 t + \sin^2 t)$$

$$= 4e^{4t} (1)$$

$$L = \int_1^4 2e^t dt = 2(e^4 - e)$$

* Gamma Function

$$\Gamma n = \int_0^{\infty} e^{-x} x^{n-1} dx \quad (n > 0)$$

$$\Gamma nu = \int_0^{\infty} e^{-x} x^n dx$$

$$\Gamma n+1 = n \Gamma n$$

$$\Gamma n = (n-1) \Gamma (n-1)$$

$$⑦ \int_0^{\pi/8} \sin^2 8x \cos^4 4x$$

$$= \frac{1}{4} \int_0^{\pi/2} \sin^2(2u) \cos^4(u) du$$

$$= \frac{1}{4} \int_0^{\pi/2} (2 \sin u \cos u)^2 \cos^4(u) du$$

$$= \frac{4}{4} \int_0^{\pi/2} \sin^2(u) \cos^6(u) du$$

$$= \int_0^{\pi/2} \sin^2 u \cos^6 u du$$

$$= \frac{1}{2} \frac{\sqrt{\frac{2+1}{2}} \sqrt{\frac{6+1}{7}}}{2 \sqrt{\frac{2+6+2}{2}}} = \frac{\sqrt{\frac{3}{2}} \sqrt{\frac{7}{2}}}{\sqrt{10}} = \frac{1}{2} \sqrt{\frac{21}{10}}$$

$$4x = u$$

$$4 dx = du$$

$$dx = \frac{du}{4}$$

	L.L	u.L
x	0	$\pi/8$
u	0	$\pi/2$

$$= \frac{\frac{1}{2} \sqrt{\frac{1}{2}} \cdot \frac{1}{2} \sqrt{\frac{1}{2}}}{2\sqrt{5}} = \frac{\frac{1}{2} \sqrt{\frac{1}{2}} \cdot \frac{3}{2} \sqrt{\frac{1}{2}}}{2 \cdot 4!}$$

$$\frac{\frac{1}{2} \sqrt{\frac{\pi}{4}} \cdot \frac{15}{8} \sqrt{\frac{\pi}{4}}}{2 \cdot 24}$$

$$\frac{15\pi}{32 \cdot 24} = \frac{15\pi}{768}$$

$$\textcircled{6} \int_0^{\pi} \sin^5 x \cos^4 x dx$$

$$\frac{1}{2} \frac{\sqrt{\frac{5+1}{2}} \sqrt{\frac{4+1}{2}}}{\sqrt{\frac{9+2}{2}}} = \frac{1}{2} \frac{\sqrt{3} \sqrt{\frac{5}{2}}}{\sqrt{\frac{11}{2}}}$$

$$= \frac{\frac{1}{2} 2! \sqrt{\frac{1}{2}}}{\frac{9}{2} \sqrt{\frac{7}{2}}} = \frac{\frac{1}{2} 2 \cdot \frac{3}{2} \sqrt{\frac{1}{2}}}{\frac{9}{2} \frac{5}{2} \frac{3}{2} \sqrt{\frac{1}{2}}} = \frac{\frac{3}{2} \sqrt{\frac{1}{2}}}{\frac{135}{8} \sqrt{\frac{1}{2}}}$$

$$= \frac{\frac{3}{2} \frac{\pi}{4}}{\frac{135}{8} \frac{\pi}{4}} = \frac{4 \frac{\pi}{4}}{45 \frac{\pi}{4}} = \frac{4\pi}{45\pi} = \frac{4}{45}$$

$$\textcircled{1} \int_0^{\infty} e^{-x^2} x^9 dx$$

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$$

Let, $x^2 = y$

$$2x dx = dy$$

=

$$\textcircled{4} \int_0^{\infty} \frac{1}{1+x^4} dx$$

$$= \int_0^{\pi/2} \frac{1}{1+\tan^2 \theta} \cdot \frac{1}{2\sqrt{\tan \theta}} \sec^2 \theta d\theta$$

$$= \int_0^{\pi/2} \frac{\sec^2 \theta}{\sec^2 \theta} \cdot \frac{1}{2\sqrt{\tan \theta}} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \sqrt{\cot \theta} d\theta = \frac{1}{2} \int_0^{\pi/2} \sqrt{\frac{\cos \theta}{\sin \theta}} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} \sin^{-1/2} \theta \left(-\frac{1}{\sin^2 \theta}\right) d\theta = \cos \frac{1}{2} \theta d\theta$$

$$= \frac{1}{2} \frac{\Gamma(1/4) \Gamma(3/4)}{\Gamma(1/4+3/4)} = \frac{1}{4} B\left(\frac{1}{4}, \frac{3}{4}\right)$$

$$(2) \int_0^b y \sqrt{b^2 - y^2} dy$$

$dy = b dz$		
y	0	b
z	0	1

$$= \int_0^1 b^5 z^5 \sqrt{b^2 - b^2 z^2} b dz$$

$$= b^6 \int_0^1 z^5 \sqrt{1 - z^2} dz$$

$$= b^6 \int_0^1 z^5 (1 - z^2)^{1/2} dz$$

$$= b^6 \int_0^1 z^4 (1 - z^2)^{1/2} dz$$

$$= b^6 \int_0^1 \frac{1}{2} x^2 (1 - x)^{1/2} dx$$

$$= \frac{b^6}{2} \int_0^1 x^{3-1} (1 - x)^{3/2-1} dx$$

$$= \frac{b^6}{2} B\left(3, 3/2\right)$$

$$= \frac{b^6}{2} \left(\frac{\Gamma(3) \Gamma(3/2)}{\Gamma(3 + 3/2)} \right) = \frac{b^6}{2} \cdot \frac{\Gamma(3) \Gamma(3/2)}{\Gamma(9/2)}$$

$$(8) \int_0^\infty \sqrt{x} e^{-x^2} dx$$

Let
 $x^2 = y$

$$2x dx = dy$$

$$dx = \frac{1}{2\sqrt{y}} dy$$

$$I = \int_0^\infty x^{-1} x^{n-1} dx$$

$$= \int_0^\infty e^{-y} y^{n-1} dy$$

$$= \int_0^\infty e^{-t} t^{n-1} dt$$

$$= \int_0^{\infty} e^{-y} (y)^{1/2} \frac{1}{2\sqrt{y}} dy$$

$$\frac{x}{y} \bigg|_0^{\infty} = \frac{\infty}{\infty} - \frac{0}{0}$$

$$= \int_0^{\infty} e^{-y} y^{1/4} \frac{1}{2\sqrt{y}} dy$$

$$= \frac{1}{2} \int_0^{\infty} e^{-y} y^{1/4} y^{-1/2} dy$$

$$= \frac{1}{2} \int_0^{\infty} e^{-y} y^{1/4 - 1/2} dy = \frac{1}{2} \int_0^{\infty} e^{-y} y^{-1/4} dy$$

$$= \frac{1}{2} \Gamma_{3/4}$$

$$\textcircled{3} \int_0^1 (1 - \frac{1}{x})^{1/3}$$

$$= \int_0^1 x^{(1/x)^0} (1 - \frac{1}{x})^{1/3}$$

$$= \frac{\Gamma(1) \Gamma(4/3)}{\Gamma(7/3)} = \frac{\frac{1}{3} \Gamma(1/3)}{\frac{4}{3} \Gamma(4/3)} = \frac{\frac{1}{3} \Gamma(1/3)}{\frac{4}{3} \frac{1}{3} \Gamma(1/3)}$$

$$= \frac{3}{4}$$

$$\int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$\textcircled{5} \int_0^1 x^{\frac{5}{2}} (1-x)^{\frac{3}{2}} dx$$

$$1) \frac{\sqrt{\frac{7}{2}} \sqrt{\frac{5}{2}}}{14} = \frac{\frac{5}{2} \sqrt{\frac{5}{2}} \frac{3}{2} \sqrt{\frac{3}{2}}}{3!}$$

$$= \frac{\frac{5}{2} \frac{3}{2} \sqrt{\frac{3}{2}} \frac{3}{2} \frac{1}{2} \sqrt{\frac{1}{2}}}{6}$$

$$2) \textcircled{1}$$

$$= \frac{45}{16} \frac{1}{2} \sqrt{\pi}$$

$$= \frac{45 \sqrt{\pi}}{32 \times 6} = \frac{15 \sqrt{\pi}}{32 \times 2}$$

$$= \frac{15}{64} \sqrt{\pi}$$

$$B(m,n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

⑤

$$y = x^{\frac{2}{3}}$$

from $x=1$
 $x=8$

6.4

$$f'(x) = \frac{2}{3} x^{-\frac{1}{3}}$$

$$1 + [f'(x)]^2 = 1 + \frac{4}{9} x^{-\frac{2}{3}} = 1 + \frac{4}{9 x^{\frac{2}{3}}}$$

$$\sqrt{\frac{9x^{\frac{2}{3}} + 4}{9x^{\frac{2}{3}}}}$$

$$L_2 = \int_1^8 (9x^{\frac{2}{3}} + 4)^{\frac{1}{2}} \left(\frac{x}{3}\right)^{\frac{1}{3}} dx$$

$$= \frac{1}{18} \int_{13}^{40} \frac{1}{2} dy$$

$$= \frac{1}{27} \left[u^{\frac{3}{2}} \right]_{13}^{40} = \frac{1}{27} (40\sqrt{40} - 13\sqrt{13})$$

$$\frac{dx}{dt} = 2 \sin 2t$$

$$\frac{dy}{dt} = 2 \cos 2t$$

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (-2 \sin 2t)^2 + (2 \cos 2t)^2$$

$$= 4 \sin^2 2t + 4 \cos^2 2t = 4 (\sin^2 2t + \cos^2 2t)$$

$$= 4$$

$$L = \int_0^{\frac{\pi}{2}} 2 dt =$$

$$= [2t]_0^{\frac{\pi}{2}} = \pi$$

$$(32) \quad x = e^t (\sin t + \cos t)$$

$$y = e^t (\cos t - \sin t)$$

$$\frac{dx}{dt} = e^t (\sin t + \cos t) + e^t (\cos t - \sin t)$$

$$= 2e^t \cos t$$

$$\frac{dy}{dt} = e^t (\cos t - \sin t) + e^t (-\sin t - \cos t)$$

$$= -2e^t \sin t$$

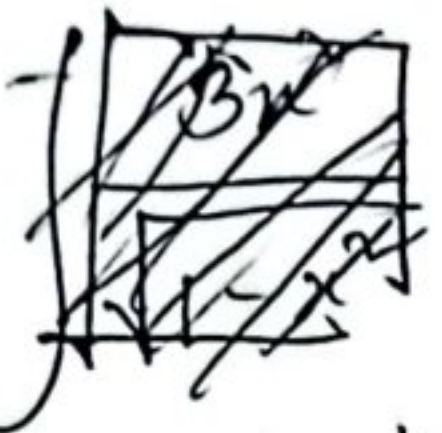
$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = (2e^t \cos t)^2 + (-2e^t \sin t)^2$$

$$= 4e^{2t} \cos^2 t + 4e^{2t} \sin^2 t$$

$$= 4e^{2t} (\cos^2 t + \sin^2 t)$$

$$L = \int_1^4 2e^t dt = 2(e^4 - e)$$

Chapter 7.5

⑨  $\int \frac{dx}{x^2 - 3x - 4} = \frac{1}{x^2 - 4x + x - 4} = \frac{1}{x(x-4) + 1(x-4)}$

$$\int \frac{dx}{x^2 - 3x - 4} \equiv \frac{A}{(x+1)} + \frac{B}{(x-4)}$$

$$1 = A(x-4) + B(x+1)$$

$$\text{Let } x = 4 \Rightarrow$$

$$1 = A(0) + B(5)$$

$$\boxed{B = 1/5}$$

$$x = -1 \Rightarrow$$

$$1 = A(-1-4) + B(0)$$

$$\boxed{A = -1/5}$$

$$\therefore \int \frac{dx}{x^2 - 3x - 4} = \frac{-1}{5(x+1)} + \frac{1}{5(x-4)}$$

$$= \int -\frac{1}{5} \ln(x+1) + \frac{1}{5} \ln(x-4)$$

$$\Rightarrow \frac{1}{5} \ln \left(\frac{x-4}{x+1} \right) + C$$

⑪ $\int \frac{11x+17}{2x^2+7x-4} = \frac{11x+7}{2x^2+8x-x-4} = \frac{11x+7}{2x(x+4)-1(x+4)}$

$$= \frac{11x+7}{(2x-1)(x+4)}$$

$$\frac{11x+17}{(2x-1)(x+4)} = \frac{A}{2x-1} + \frac{B}{x+4}$$

$$11x+17 = A(x+4) + B(2x-1)$$

$$\begin{aligned} x &= -4 \\ 11(-4)+17 &= A(0) + B(-8-1) \\ -44+17 &= -9B \end{aligned}$$

$$\begin{aligned} 11x+17 &= Ax+4A+2Bx-B \\ &= x(A+2B)+4A-B \end{aligned}$$

$$A+2B=11 \quad \text{--- (1)}$$

$$4A-B=17 \quad \text{--- (2)}$$

$$\begin{aligned} \text{(1)} \times 2 & \Rightarrow A=5 \\ B &= 3 \end{aligned}$$

$$\therefore \int \frac{5}{2x-1} + \int \frac{3}{x+4} = 5 \ln|2x-1| + 3 \ln|x+4|$$

$$\Rightarrow \frac{5}{2} \int \frac{du}{u} + 3 \ln|x+4|$$

$$= \frac{5}{2} \ln|2x-1| + 3 \ln|x+4|$$

$$\begin{aligned} 2x-1 &= u \\ 2dx &= du \\ dx &= \frac{du}{2} \end{aligned}$$

$$\textcircled{3} \int_0^{\infty} e^{-2n} dn$$

6.5

Q6 $y = x^3$ $0 \leq x \leq 1$

$$g(x) = x^3$$

$$g' = 3x^2$$

$$S = \int_0^1 2\pi x^3 \sqrt{1 + (3x^2)^2} dx$$

$$S = 2\pi \int_0^1 (1 + 4x^4)^{\frac{1}{2}} x^3 dx$$

$$= \frac{\pi}{18} \int_0^1 \frac{(1 + 9x^4)^{\frac{3}{2}}}{\frac{1}{2} + 1} dx$$

$$= \frac{\pi}{18} \cdot \frac{2}{3} \left[(1 + 9x^4)^{\frac{3}{2}} \right]_0^1$$

$$= \frac{\pi}{27} \left[(1 + 9)^{\frac{3}{2}} - (1 + 0)^{\frac{3}{2}} \right] = \frac{\pi}{27} (10^{\frac{3}{2}} - 1)$$

Q8 $x = 2\sqrt{1-y}$

$$g(x) = 2\sqrt{1-y}$$

$$= 2(1-y)^{\frac{1}{2}}$$

$$g'(x) = 2 \left(\frac{1}{2\sqrt{1-y}} \right) \frac{d}{dx} (1-y)$$

$$g'(x) = \frac{1}{\sqrt{1-y}} \cdot \frac{dy}{dx} (-1) = -\frac{1}{\sqrt{1-y}}$$

$$S = \int_{-1}^0 2\pi \cdot 2\sqrt{1-y} \cdot \sqrt{1 + \left(\frac{-1}{\sqrt{1-y}} \right)^2} dy$$

$$= 4\pi \int_{-1}^0 \sqrt{1-y} \sqrt{\frac{1+y-1}{1-y}}$$

$$= 4\pi \int_{-1}^0 \sqrt{1-y} \frac{\sqrt{2-y}}{\sqrt{1-y}}$$

$$\Rightarrow 4\pi \int_{-1}^0 (2-y)^{1/2}$$

$$= 4\pi \left[\frac{2}{3} (2-y)^{3/2} \right]_{-1}^0$$

$$= \frac{8\pi}{3} \left[(2-0)^{3/2} - (2+1)^{3/2} \right] = \frac{8\pi}{3} [2\sqrt{2} - 3\sqrt{3}]$$

$$= \frac{8\pi}{3} [2\sqrt{2} - 3\sqrt{3}]$$

$$(4) \int_0^{\infty} \frac{1}{1+x^4} dx$$

$$= \int_1^{\infty} + \frac{(1+x^4)^2}{-4x^3}$$

$$\Rightarrow \frac{1}{4} \int_0^1 t \times \frac{\frac{1}{t}}{\left(\frac{1-t}{t}\right)^{3/4}} dt$$

$$= \frac{1}{4} \int_0^1 \frac{1}{t} \times t^{3/4} (1-t)^{-3/4} dt$$

$$= \frac{1}{4} \int_0^1 t^{3/4-1} (1-t)^{1/4-1} dt$$

$$= \frac{1}{4} B\left(\frac{3}{4}, \frac{1}{4}\right)$$

$$= \frac{1}{4} \times \frac{\Gamma_{3/4} \Gamma_{1/4}}{\Gamma_1}$$

$$= \frac{\Gamma_{3/4} \Gamma_{1/4}}{4}$$

$$\text{Let } \frac{1}{1+x^4} = t$$

$$(1+x^4)^{-1} = t$$

$$\frac{-4x^3}{(1+x^4)^2} = \frac{dt}{dx}$$

$$dx = \frac{(1+x^4)^2}{-4x^3} dt$$

$$(6) \int_0^{\pi} \sin^5 \theta \cos^4 \theta d\theta$$

$$= \frac{\sqrt{\frac{5+1}{2}} \sqrt{\frac{4+1}{2}}}{\sqrt{\frac{5+4+2}{2}}}$$

$$= \frac{\sqrt{3} \sqrt{\frac{5}{2}}}{\sqrt{\frac{1}{2}}} = \frac{2! \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\frac{1}{2}}}{\frac{9}{2} \cdot \frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\frac{1}{2}}}$$

$$= \frac{16}{315}$$